

Regression Algorithms for Learning & Its Application on Cryptocurrency Markets

Professional Issues Report

Syed Maesum Hussain Zaidi

Undergraduate Final Year Project

BSc Computer Science (Artificial Intelligence)



Supervised By : Xu Feng

Department of Computer Science

Royal Holloway, University of London

Declaration

This report has been prepared on the basis of my own work. Where other published and unpublished source materials have been used, these have been acknowledged.

Word Count: 903

Student Name: Syed Maesum Hussain Zaidi

Date of Submission: 15 April, 2026

Signature:

A handwritten signature in black ink that reads "Maesum" followed by a long, sweeping diagonal stroke.

Professional Issues in Cryptocurrency Price Prediction System

Professional issues in computing are concerned with the ethical, social, and practical implications of developing and deploying software systems. In the case of this project which focuses on predicting cryptocurrency prices using regression based and other machine learning algorithms, there are multiple professional issues that arise. These include reliability and trust, ethical responsibility, data usage and privacy, and the potential impact of such systems on users.

One of the main professional issues in this project is reliability and trust. The system developed in this project attempts to predict future cryptocurrency prices based on historical data. However, financial markets and in particular cryptocurrency markets are highly volatile and unpredictable. Even though the models implemented may show good performance on historical data, there is no guarantee that they will perform well in real world conditions. This creates a risk that users may place too much trust in the predictions produced by the system. Hence, it could lead to significant financial loss if a user were to rely on these predictions for financial decisions. As a result, it is important to clearly communicate the limitations of the model and ensure that users understand that the predictions are not guaranteed.

Another separate issue which is closely related to this is ethical responsibility. As a developer, there is a responsibility to consider how the system may be used and the potential consequences of its use. The system for this project is designed for educational and analytical purposes rather than for real financial decision making. However, it would still be possible for users to misuse the system by treating it as financial advice. This raises the question of whether additional safeguards or warnings should be implemented. An example of this would be the use of disclaimers or limiting how predictions are presented. This reflects the broader responsibility of computing professionals to consider the societal impact of their work. It is also highlighted by organisations such as the British Computer Society (BCS) and the Association for Computing Machinery (ACM).

Data usage and Privacy is another important issue which is relevant to this project. The system relies on historical cryptocurrency data obtained from Yahoo Finance through the yfinance library. While the data itself is publicly available and does not involve personal information, it still raises considerations about the ethical use of data. It is important to ensure that data is used in accordance with its intended purpose. This project does not involve personal data however many real world machine learning systems do. Therefore, developers must be aware of privacy concerns such as data protection and compliance with regulations like GDPR.

The issue of model bias and limitations is also relevant. The models used in this project are trained purely on historical price data and do not take into account external factors such as news events, market sentiment or macroeconomic conditions. This means that the model has an incomplete view of the factors that influence cryptocurrency prices. Hence, the predictions may be biased or misleading in certain situations. This shows a key professional issue in machine learning which is that models are only as good as the data they are trained on. Developers must be careful not to overstate the capabilities of their models and should clearly communicate any limitations.

There are also usability and accessibility considerations. The project includes a graphical user interface (GUI) to make the system easier to use. While this improves usability, it also raises questions about how information is presented to users. For example, outputs such as prediction graphs and performance metrics must be displayed in a way that is clear and understandable. Users may misinterpret the results if the GUI is confusing. Making sure that the system is accessible and easy to use is important as it affects how effectively users can interact with the system.

Accountability is also important as in real world applications if the system provides incorrect or misleading information, it is an important consideration to know who would be held responsible. Even though in this project the system is not deployed commercially, it still demonstrates how responsibility could become a concern. In case such a system was used in a financial context, developers, users and organizations would all hold some level of responsibility. This highlights the importance of clear documentation and transparency in how the system works.

Finally, the development process itself raises software engineering and management issues. Throughout the project, decisions had to be made regarding model selection, feature engineering, and validation methods. These decisions involved trade-offs between complexity, performance, and interpretability. Managing these trade-offs is a key part of professional practice in computing. The project also required time management, debugging, and iterative improvement, all of which are important professional skills. An example where careful problem solving was required would be making sure that the system worked consistently across different environments and handling issues such as file paths and data availability.

In conclusion, this project highlights several important professional issues in computing, particularly in the context of machine learning and financial prediction systems. These include reliability and trust, ethical responsibility, data usage, model limitations, usability, accountability, and software engineering practices.

Addressing these issues is essential for developing systems that are not only technically effective but also responsible and trustworthy. As the role of computing in today's society grows it is important for all developers to consider these broader implications alongside the technical aspects of their work.